Exercise 4.

1. SQL JOINS

1. SELECT e.EmployeeID, e.FirstName, e.LastName,
e.Department, e.salary, p.ProjectID,
p.ProjectName, p.Budget, p.Status
FROM Employees e
INNER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

2. SELECT
e.EmployeeID, e.FirstName, e.LastName,
e.Department, e.Salary, p.ProjectID,
p.ProjectName, p.Budget, p.Status
FROM Employeese
LEFT JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

3. SELECT
p.ProjectID, p.ProjectName, p.Budget,
p.Status, e.EmployeeID, e.FirstName,
e.LastName, e.Department, e.Salary
FROM Employees e
Right -

3. SELECT
p.Project, p.ProjectName, p.Budget,
p.Status, e.EmployeeID, e.FirstName,
e.LastName, e.Department, e.Salary
FROM Employees e
RIGHT JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

4. SELECT
e.EmployeeID, e.FirstName, e.LastName,
e.Department, e.Salary, p.ProjectID,
p.ProjectName, p.Budget, p.Status
FROM Employeese
FULL OUTER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

2. UNION & UNION ALL

5. SELECT City AS Location FROM Employees
UNION
SELECT Status FROM Projects;

6. SELECT City AS Location FROM Employees
UNION ALL
SELECT Status FROM Projects;

3. Filtering Statements

7. SELECT EmployeeID, FirstName LastName,
Department, Salary
FROM Employees
WHERE Salary > 70000;

8. SELECT
EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

9. SELECT
ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Status <> 'completed';

10. SELECT
ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 AND
Status <> 'Completed';

11. SELECT EmployeeID, FirstName, LastName,
Department, Salary, City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;

12. SELECT
EmployeeID, FirstName, LastName,
Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

~~13.~~ 4. Aggregate Functions

13. SELECT
Department, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;

14. SELECT City, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65000;

15. SELECT Department, COUNT(EmployeeID) AS
EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) > 1;

16. SELECT Status, COUNT(ProjectID) AS ProjectCount
FROM Projects
GROUP BY Status
HAVING COUNT(ProjectID) >= 2;

17. SELECT
e.EmployeeID, e.FirstName, e.LastName,
SUM(p.Budget) AS TotalProjectBudget
FROM Employees e
JOIN Projects p ON e.EmployeeID = p.EmployeeID
GROUP BY e.EmployeeID, ~~e.Employee~~ e.FirstName, e.LastName
HAVING SUM(p.Budget) > 150000;